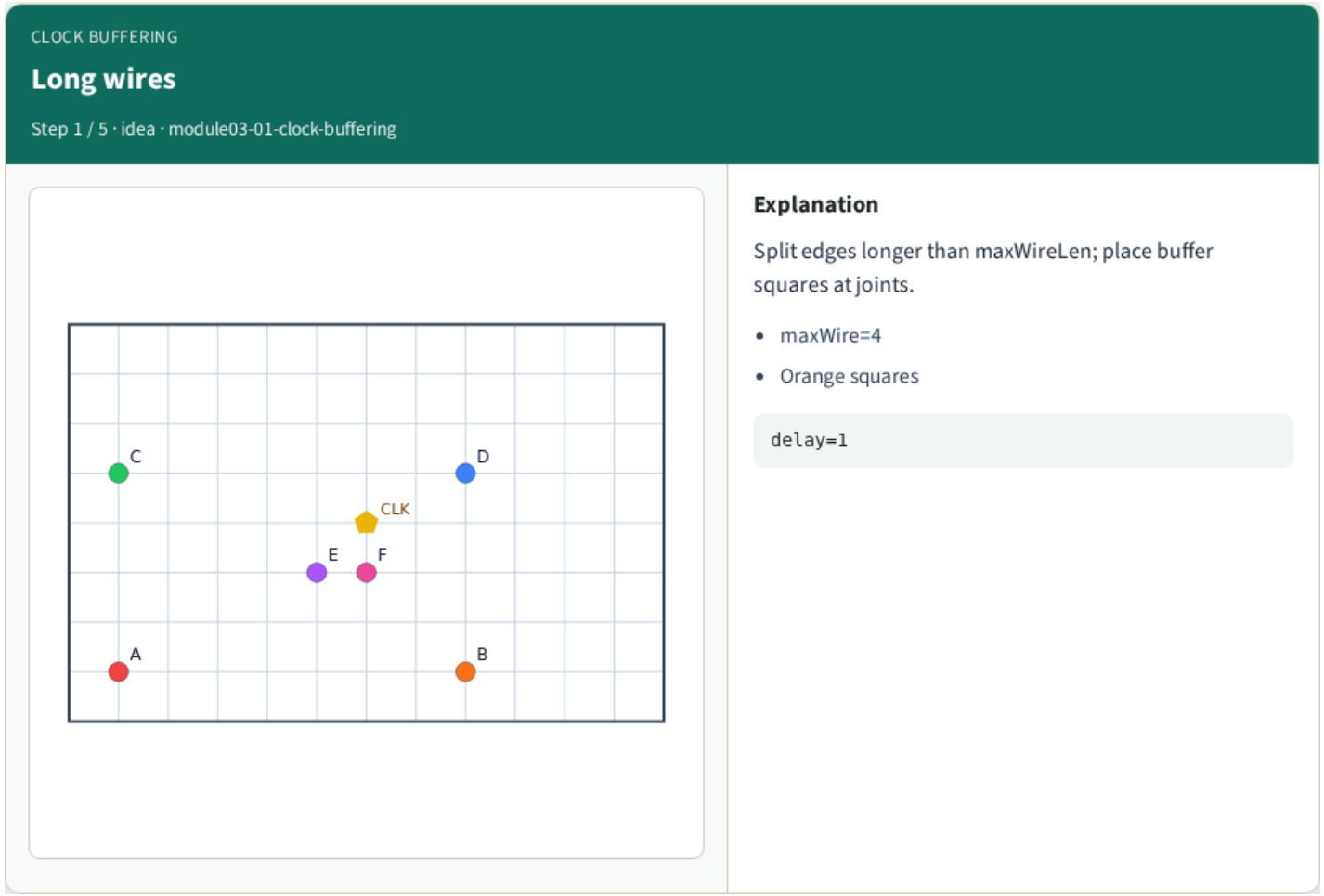
Clock buffering on wires

Long wires lose slew; CTS inserts buffers along the tree

The idea

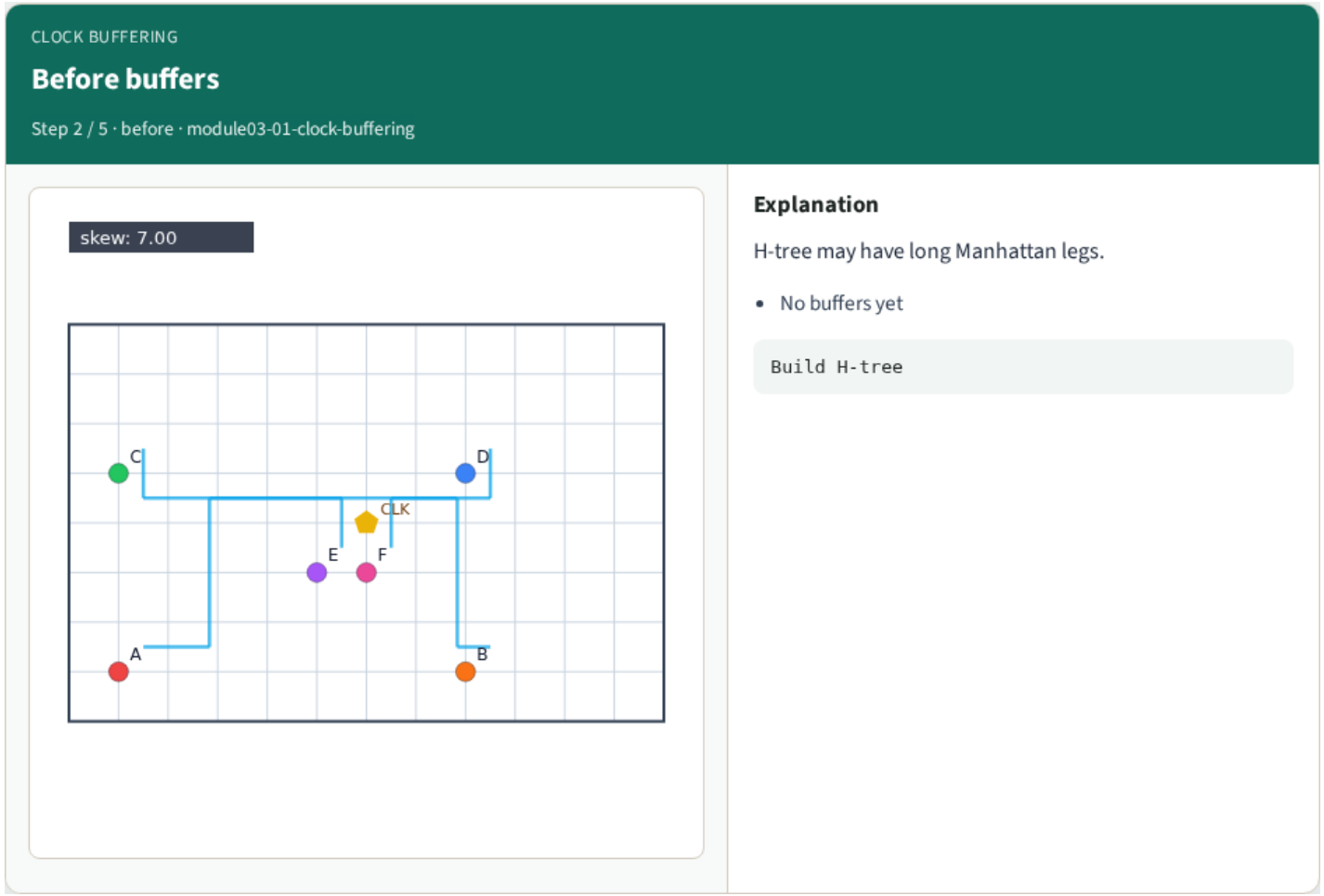
- If an edge longer than maxWireLen
- Buffered latency adds bufferDelay per buffer on the path

Long wires



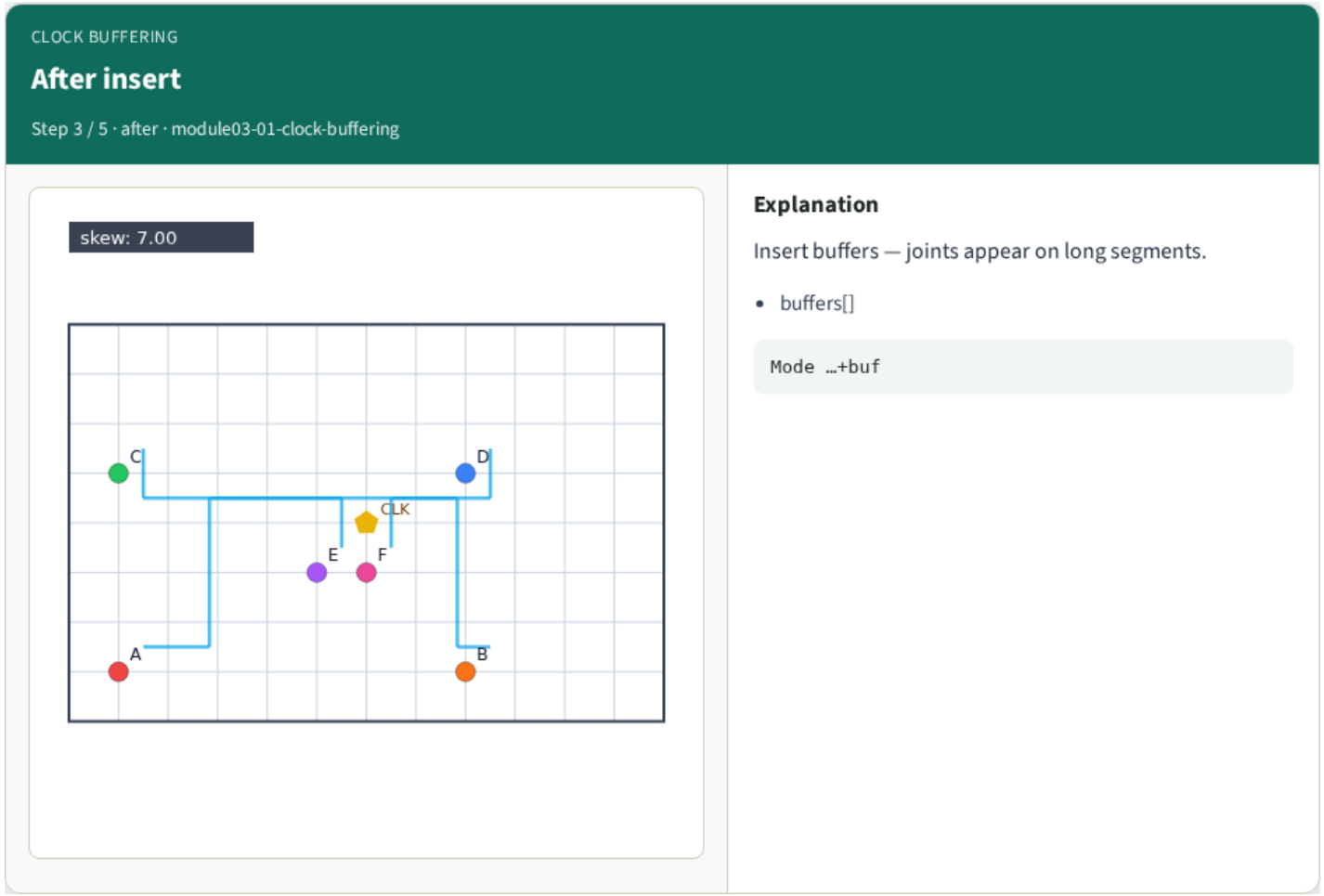
Long wires

Before buffers



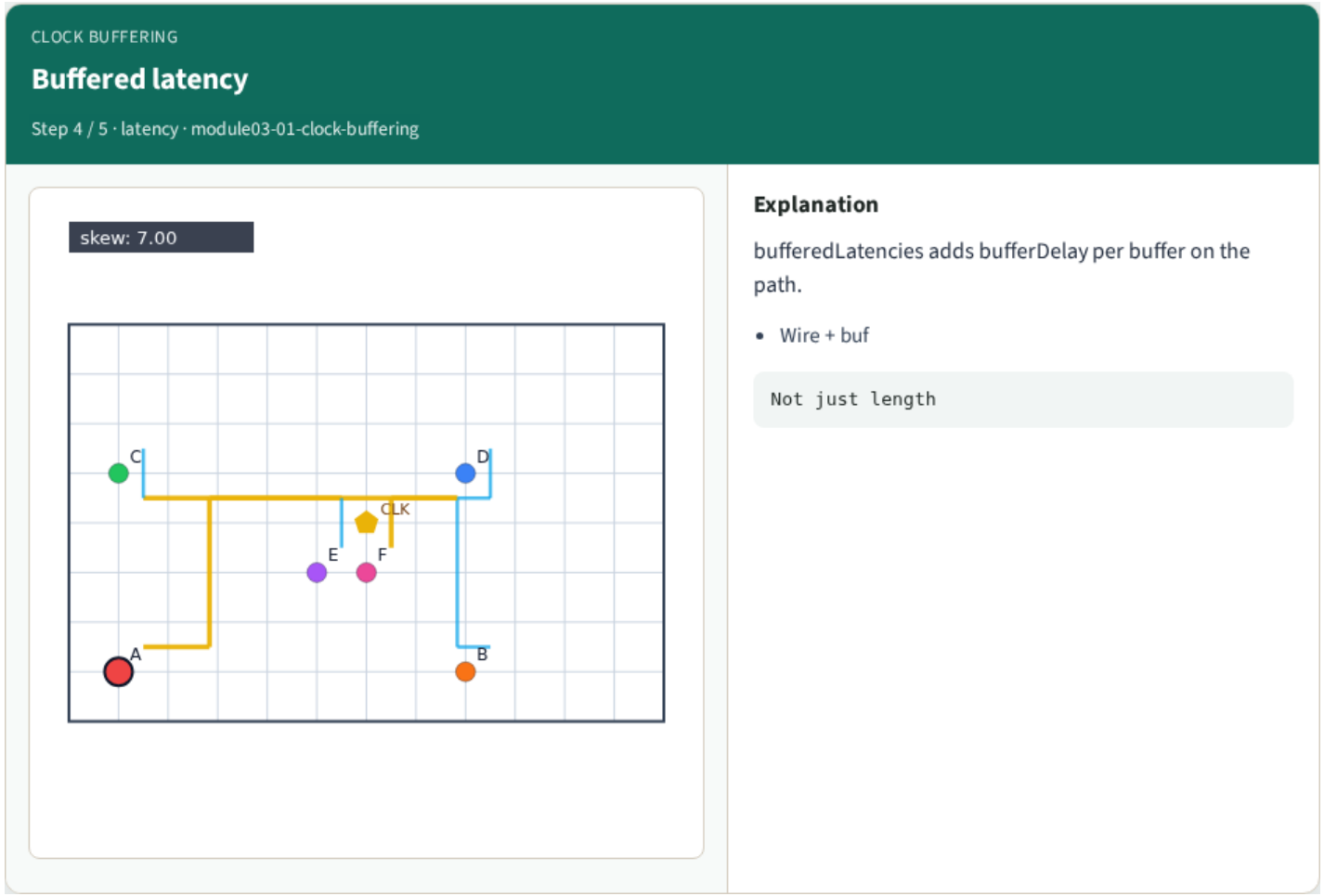
Before buffers

After insert



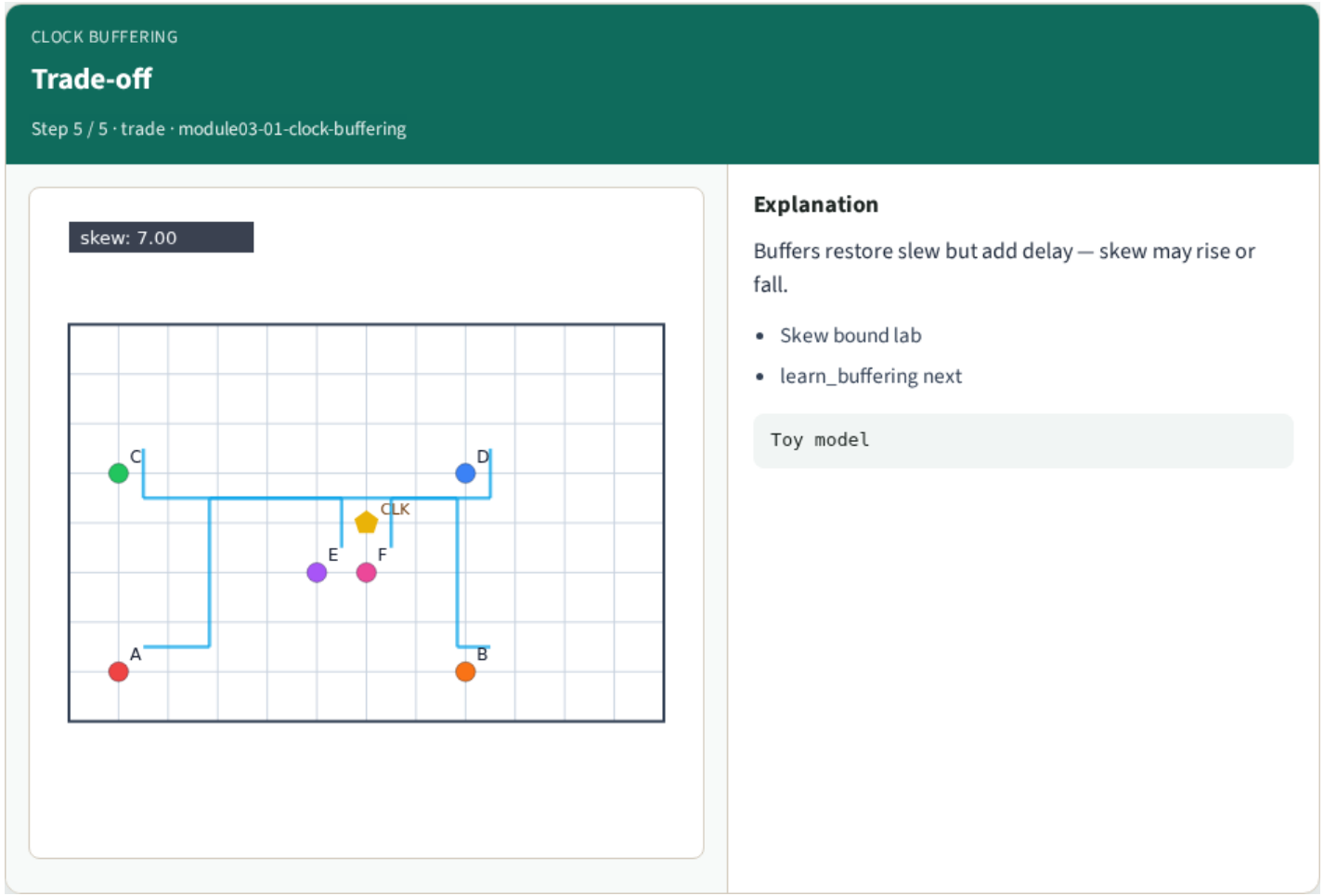
After insert

Buffered latency



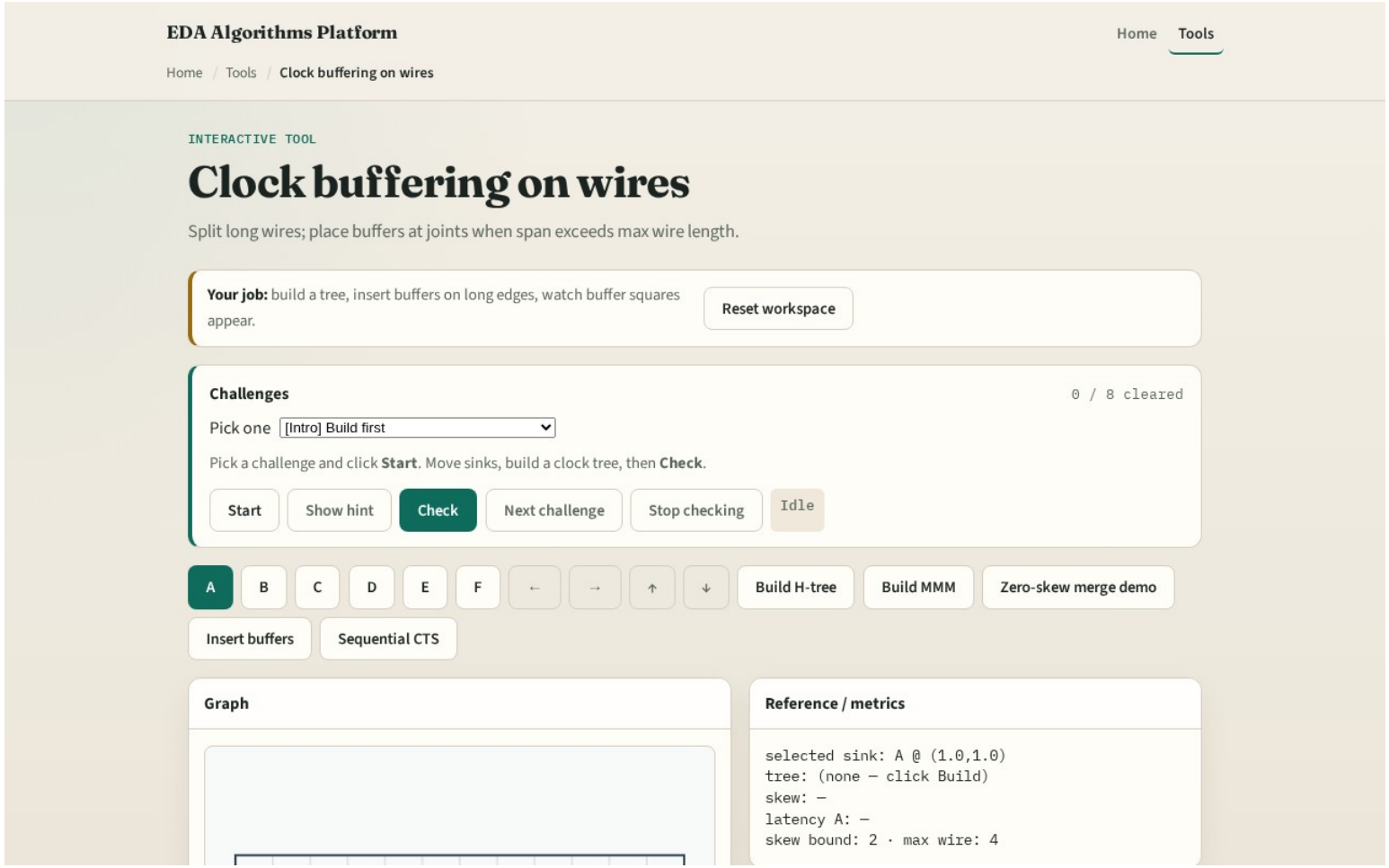
Buffered latency

Trade-off



Trade-off

Browser lab track



Browser lab starter

Implement track

- Call `insert_buffers(edge, max_span, buffer_delay)`
- Assert max segment length ≤ max_span

Pitfalls

- Buffering without splitting edges
- Counting buffer delay twice
- Using max span zero

Your turn

- Next: meet a skew bound with buffers and topology choice